TD(λ) and Eligibility Traces

Master Degree in Control Systems Engineering 2024-2025
University of Padova

Teachers: Prof. Ruggero Carli & Prof. Gian Antonio Susto

Model-free prediction : quick review

- **Goal :** estimate the value function v_π using experience

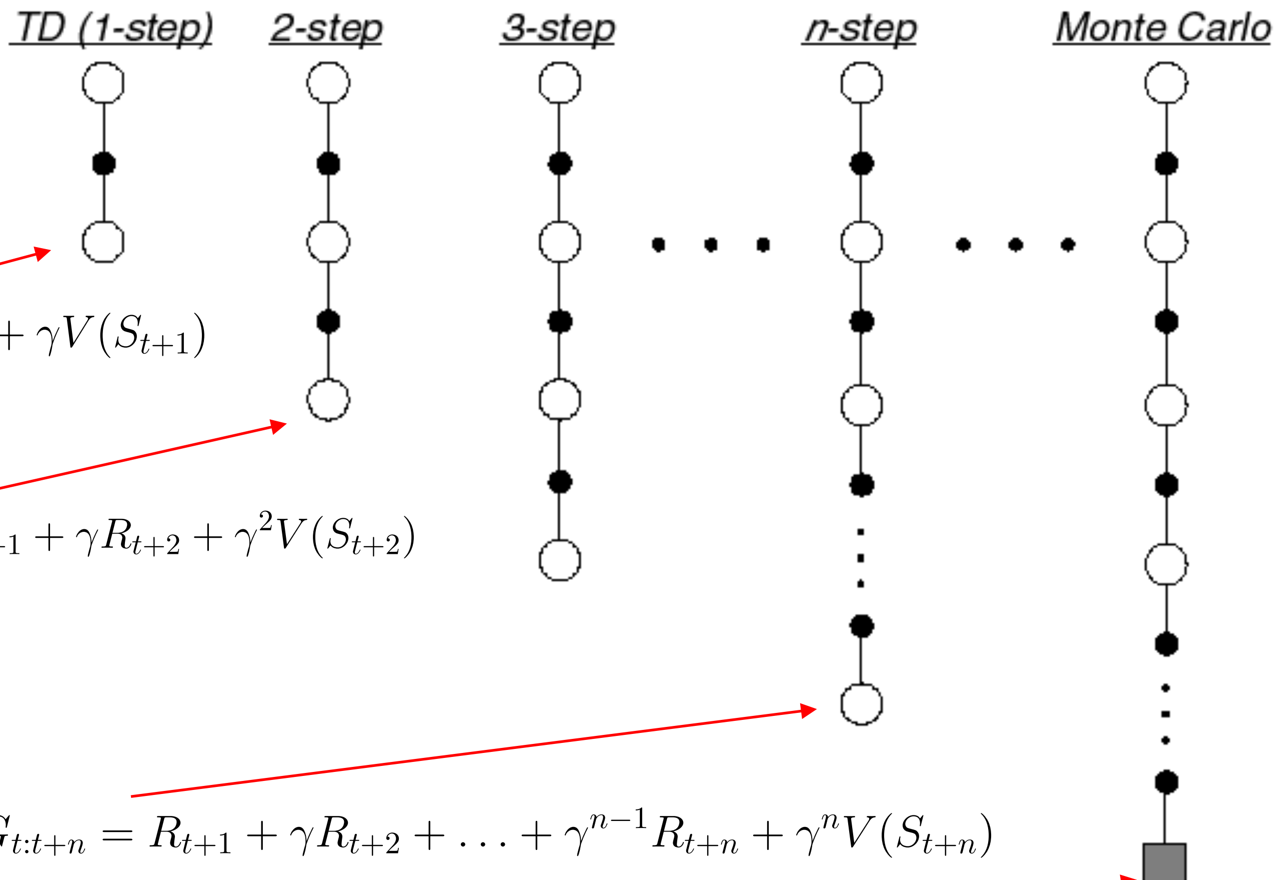
$$v_\pi(s) = \mathbb{E}_\pi \left[G_t | S_t = s \right]$$

- **Monte Carlo** methods update with complete returns
  - Observe : $G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T$
  - Update : $V(S_t) \leftarrow V(S_t) + \alpha \left[\underbrace{G_t}_{\text{MC target}} - V(S_t) \right]$

- **Temporal Difference methods** update with single rewards:
  - Observe : R_{t+1}
  - Estimate return : $G_t \approx R_{t+1} + \gamma v_\pi(S_{t+1}) \approx R_{t+1} + \gamma V(S_{t+1})$
  - Update : $V(S_t) \leftarrow V(S_t) + \alpha \left[\underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{TD target}} - V(S_t) \right]$

n-Step Prediction (extension of 1-step TD)

- Let TD target look *n* steps into the future



$$G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1})$$

$$G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$$

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T$$

n-Step Return

- Consider the following n-step returns for $n = 1, 2, \infty$:

$$
\begin{array}{lll}
n = 1 & (TD) & G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1}) \\
n = 2 & & G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2}) \\
\vdots & & \\
n = \infty & (MC) & G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T
\end{array}
$$

- Define the n-step return

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

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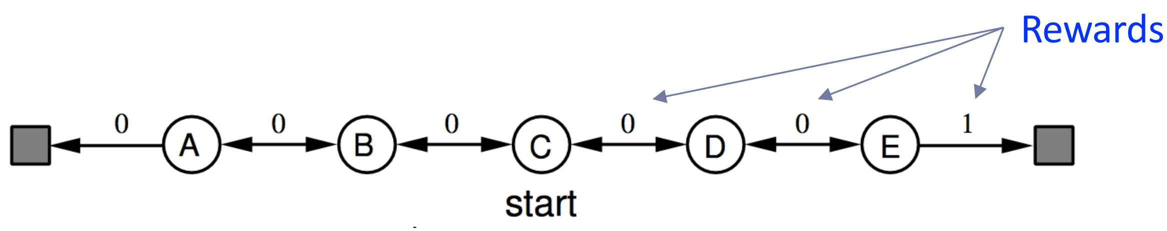
- *n*-step temporal-difference learning

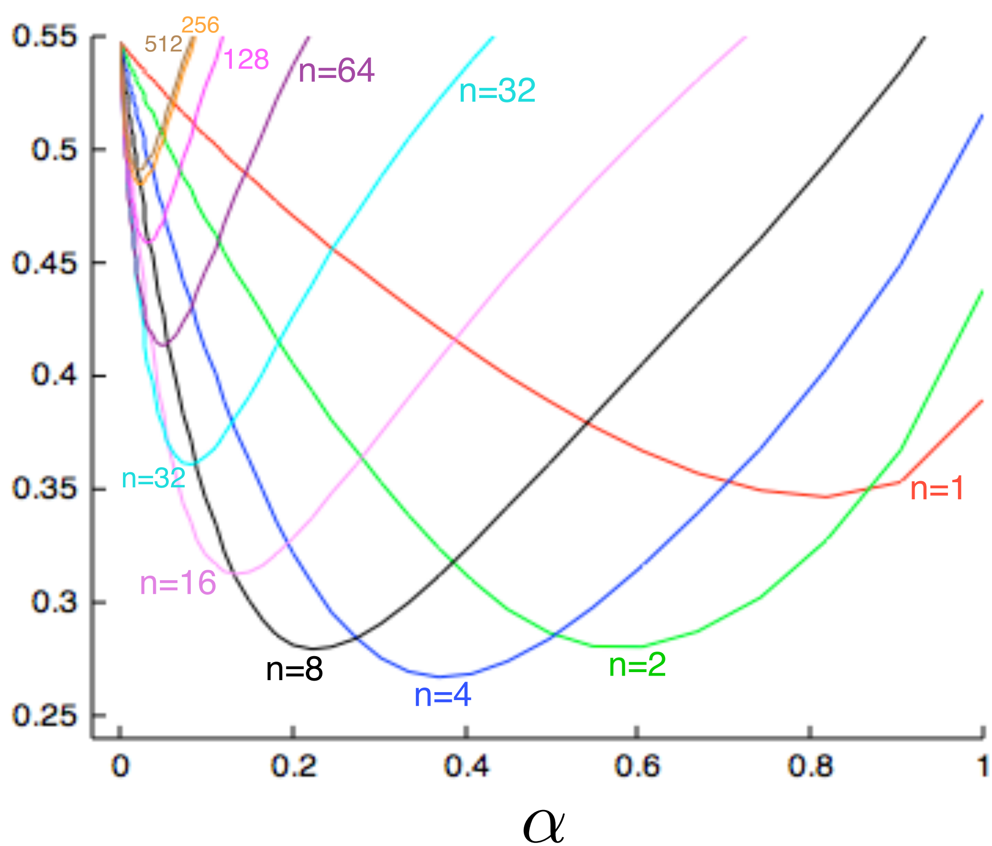
$$V(S_t) \leftarrow V(S_t) + \alpha(G_{t:t+n} - V(S_t))$$

Large random walk example: which value of n is better?

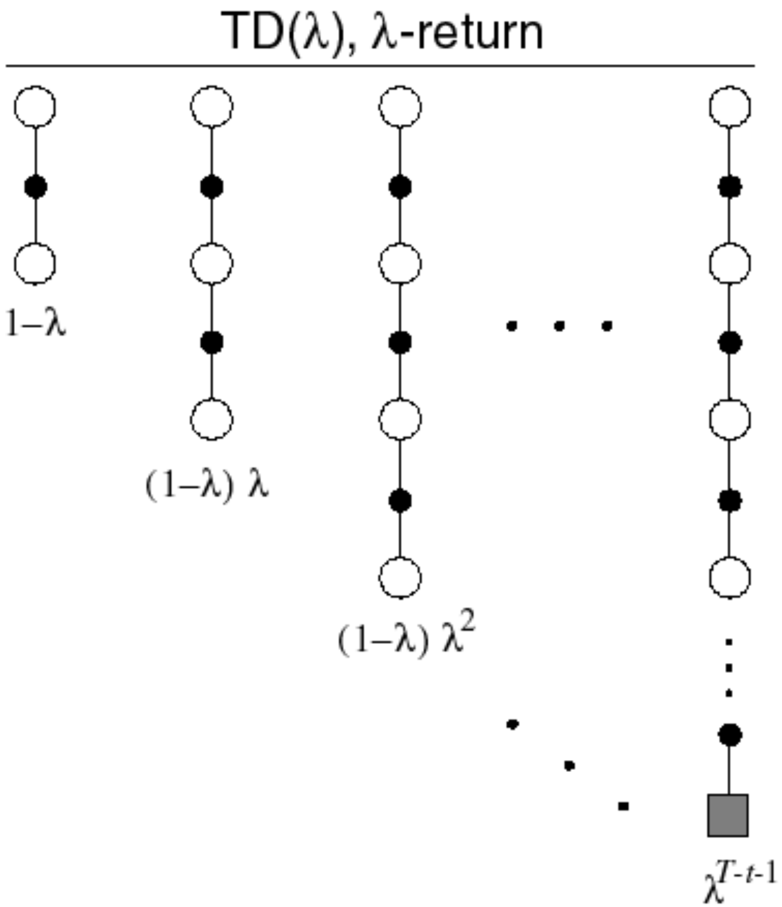
The policy selects right with p=0.5 and left with p=0.5



Average RMS error over 19 states and first 10 episodes

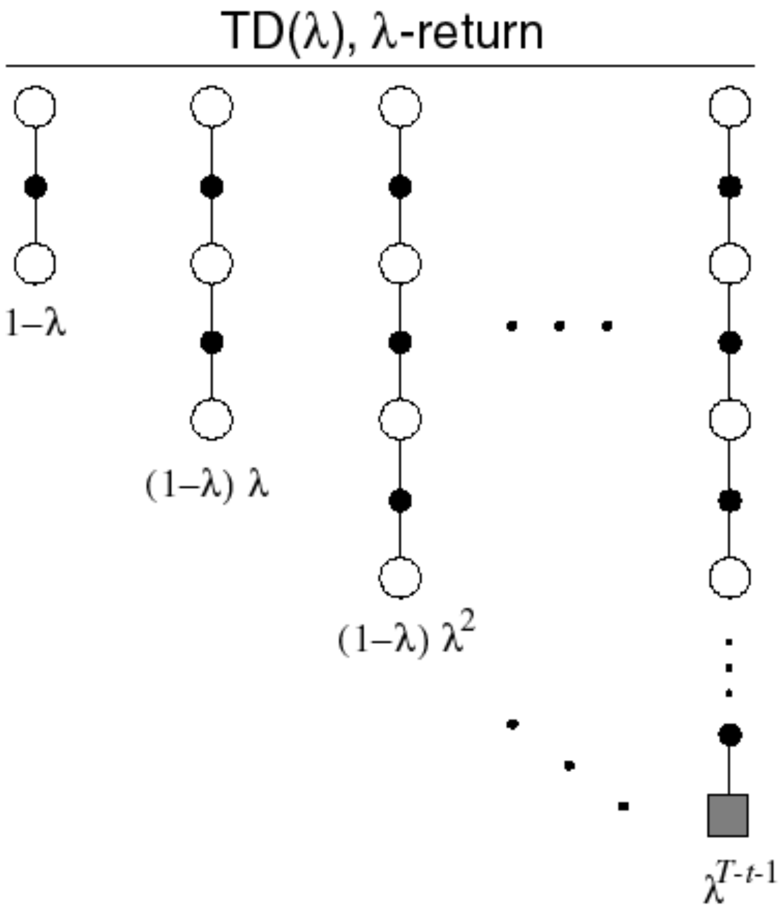


λ – return



- The λ-*return* G_t^λ combines all *n*-step returns $G_{t:t+n}$

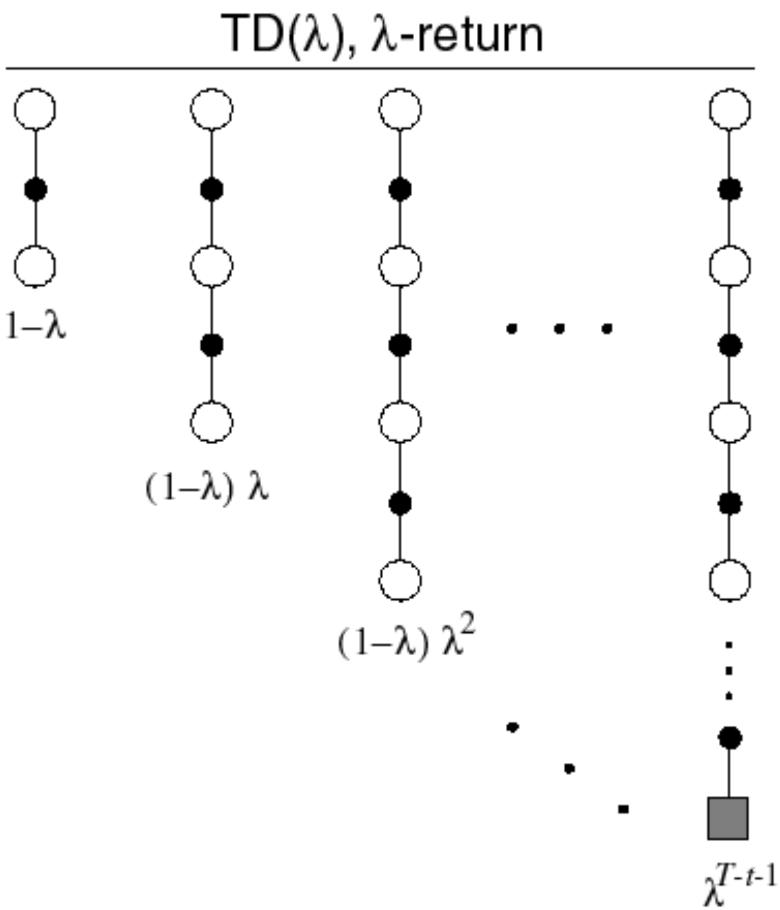
λ – return



- The λ-*return* G_t^λ combines all *n*-step returns $G_{t:t+n}$
- Using weight $(1-\lambda)\lambda^{n-1}$

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}\,G_{t:t+n}$$

λ – return



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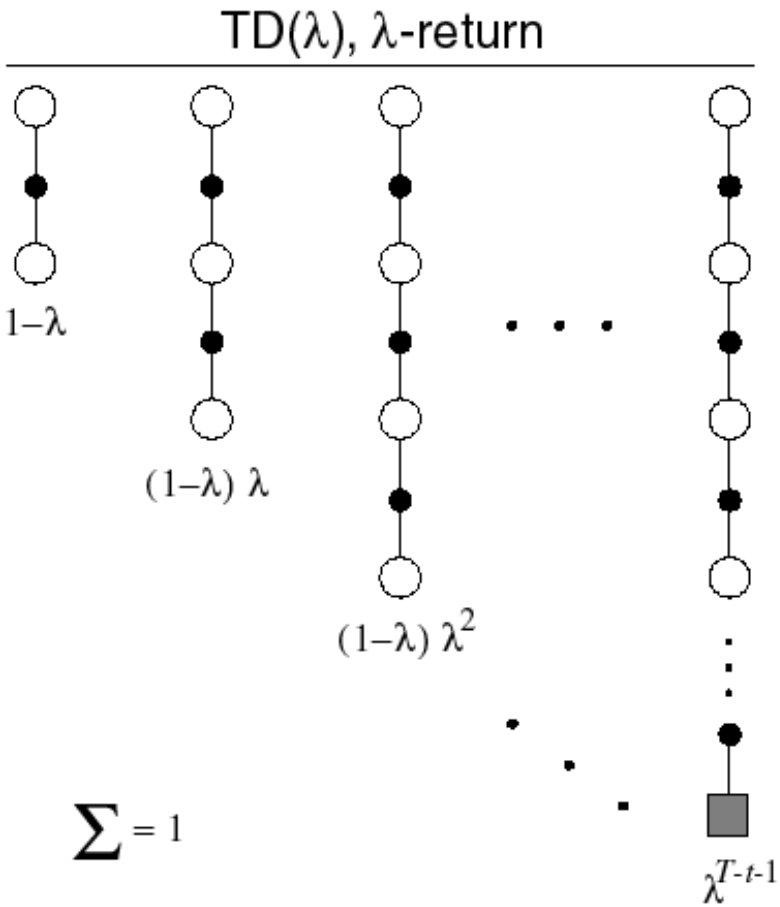
- Forward-view TD(λ)

$$V(S_t) \leftarrow V(S_t) + \alpha\left(G_t^\lambda - V(S_t)\right)$$

λ – return



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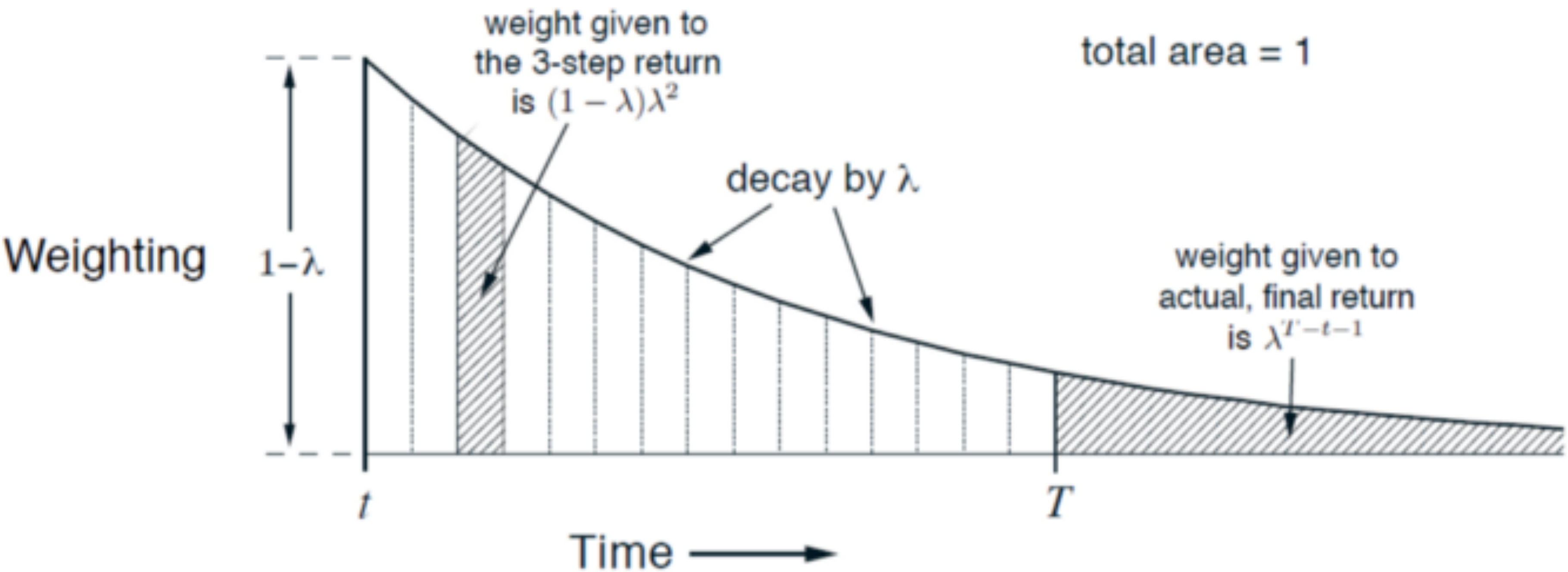
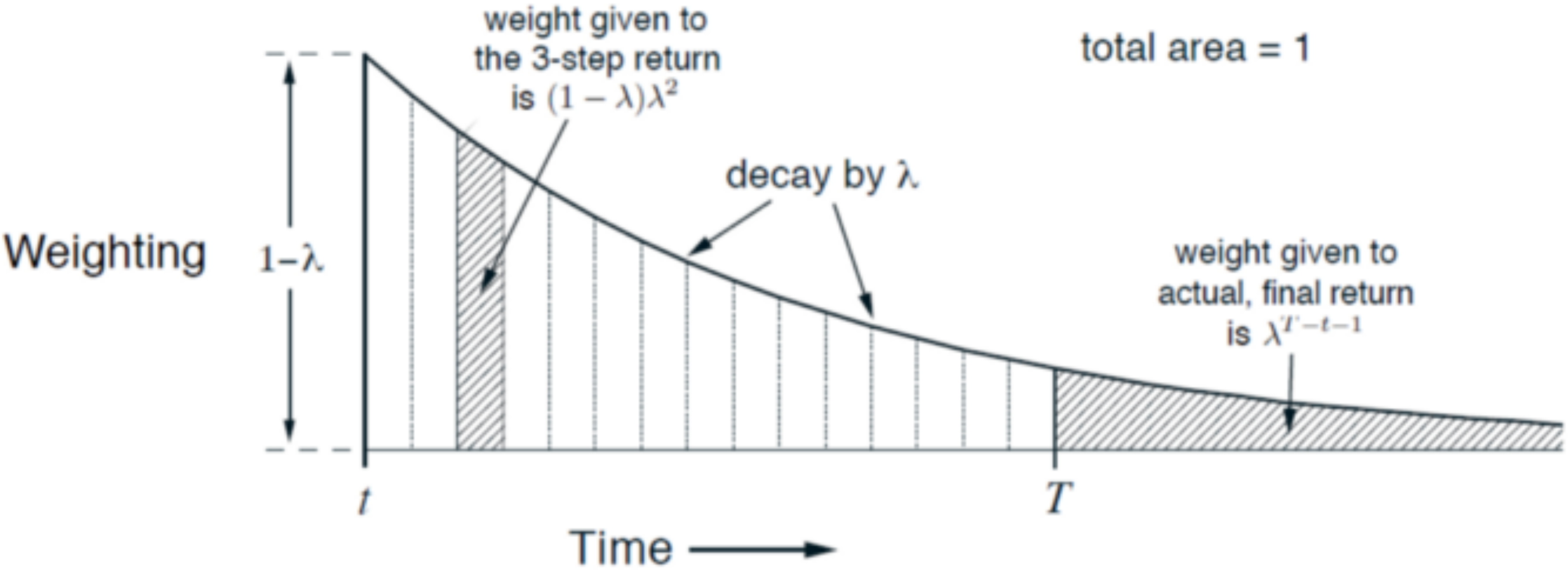
Convex combination of n-step returns (sum of weights = 1)

TD(λ) Weighting Function



TD(λ) Weighting Function

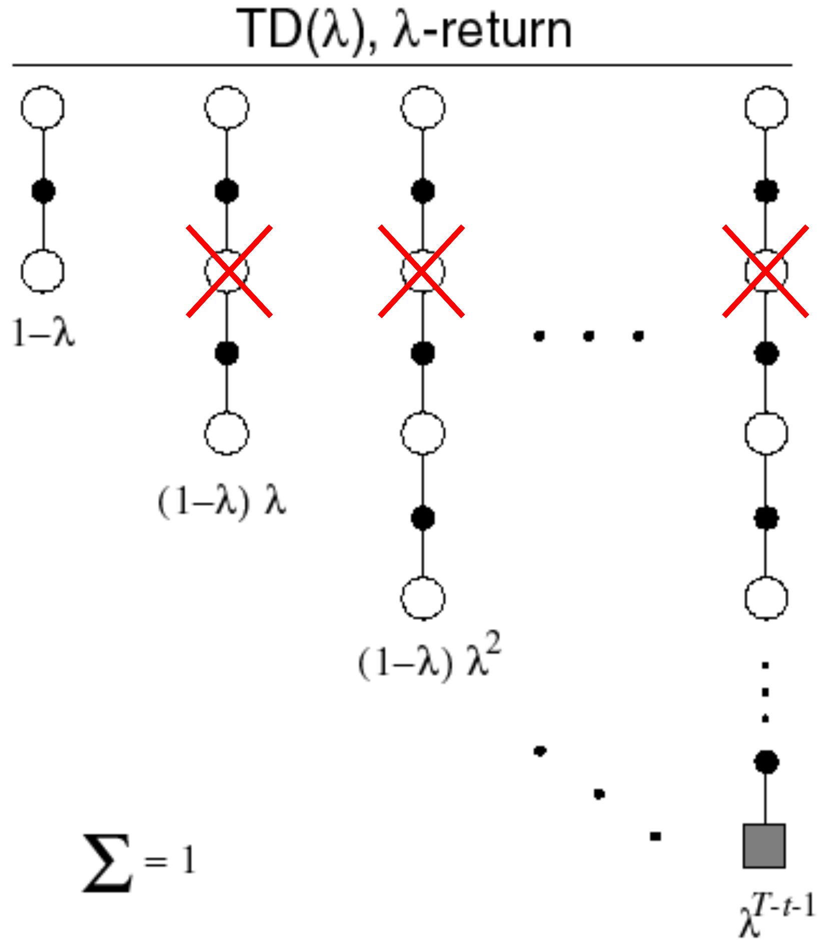


What about if an episode terminates at T?

$$G_t^\lambda = \underbrace{(1-\lambda)\sum_{n=1}^{T-t-1}\lambda^{n-1}G_{t:t+n}}_{\text{Until termination}} + \underbrace{\lambda^{T-t-1}G_t}_{\text{After termination}}$$

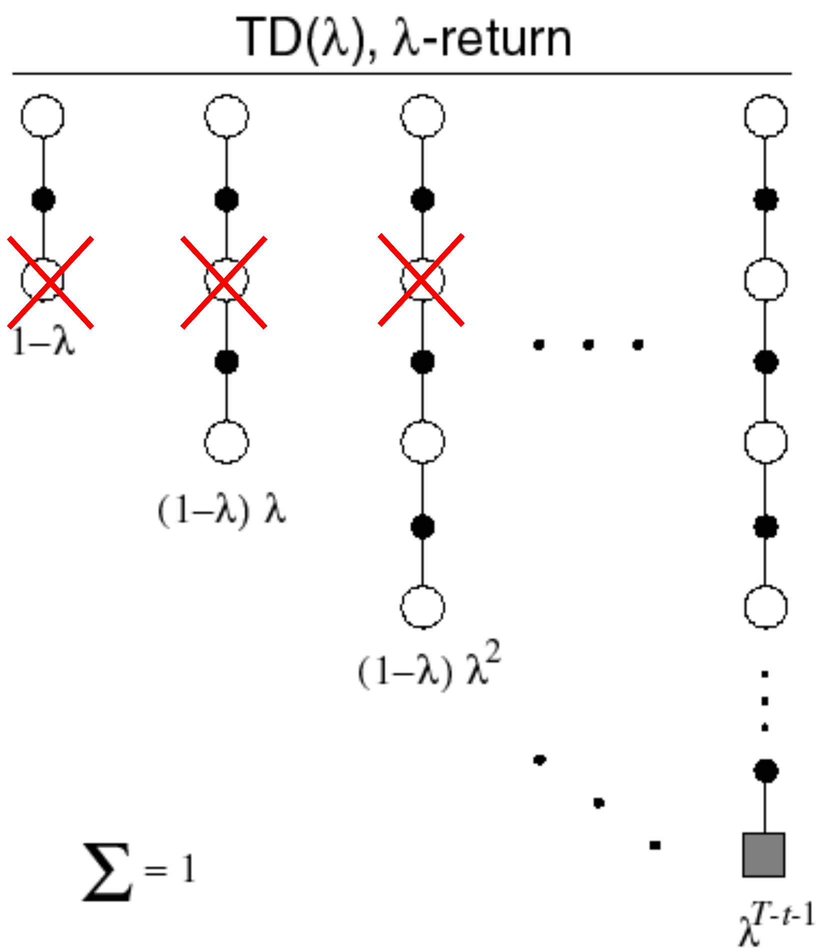
Relation to TD(0) and MC

If $\lambda = 0$ you get TD(0)

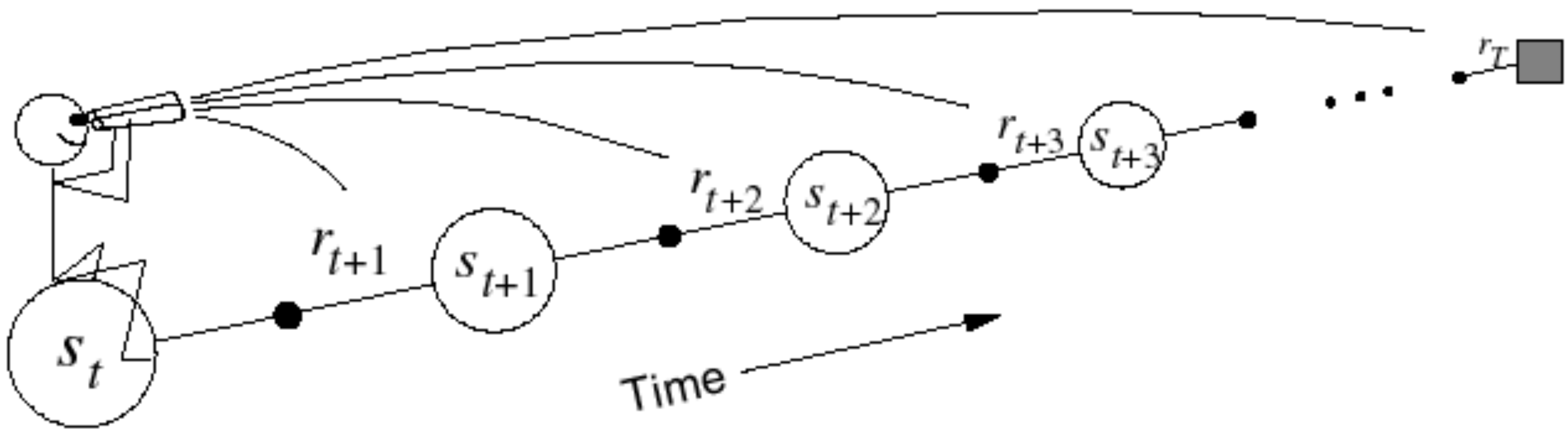


Relation to TD(0) and MC

If $\lambda = 1$ you get MC



Forward - view TD(λ)



- Update value function towards the λ-return
- Forward-view looks into the future to compute G_t^{λ}
- Like MC, can only be computed from complete episodes

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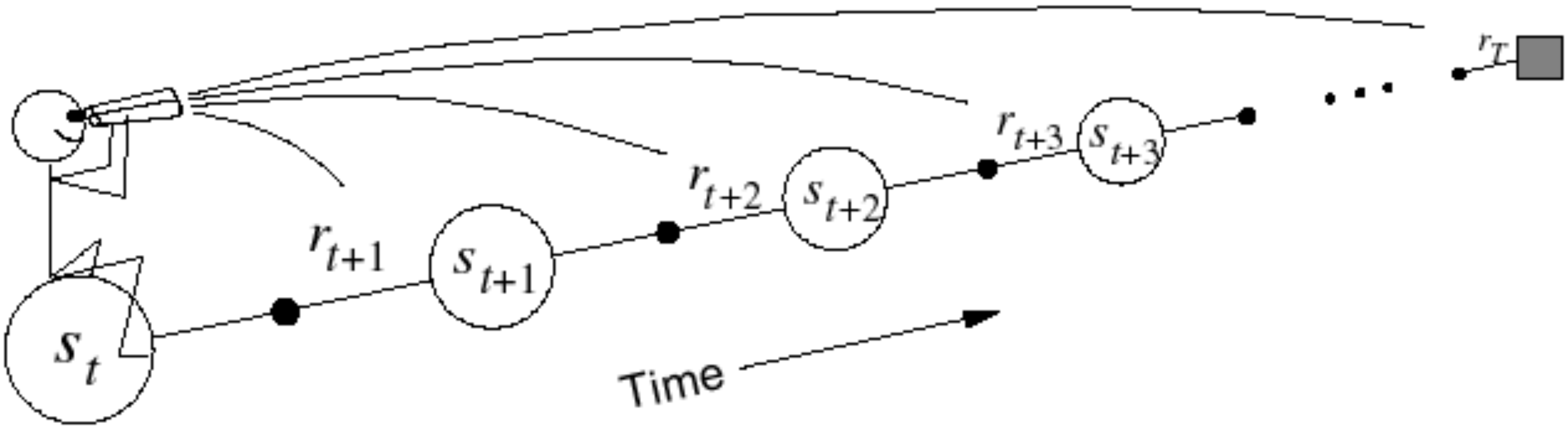

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But so far we have been doing just that! We are looking n-steps ahead…

Let us change point of view by adopting a new approach: The Backward View

Eligibility traces

Backward - view TD(λ) : eligibility traces

- Additional memory variable associated with each state, its **eligibility trace**
- $E_t(s)$ eligibility trace of state s at time t (≥ 0 !)

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- Additional memory variable associated with each state, its **eligibility trace**
- $E_t(s)$ eligibility trace of state s at time t (≥ 0 !)
- On each step, the eligibility traces of all non-visited states decay by $\gamma\lambda$

$$E_t(s) = \gamma\lambda E_{t-1}(s), \qquad \forall s \in \mathcal{S}, s \neq S_t$$

γ : *discount rate*

λ : *trace-decay parameter*

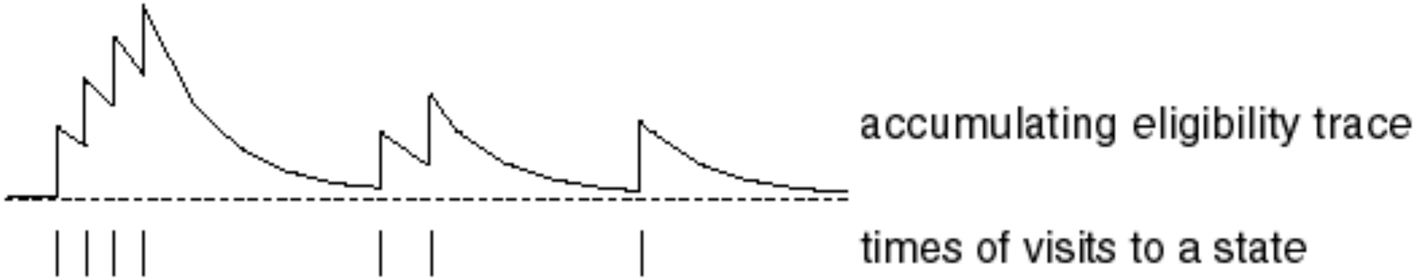
- What about the trace for S_t, the one visited at time t?

$$E_t(S_t) = \gamma\lambda E_{t-1}(S_t) + 1$$

The eligibility trace for S_t decays just like for any state, but it is incremented by 1!

Backward - view TD(λ) : eligibility traces

$$E_0(s) = 0$$
$$E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$$



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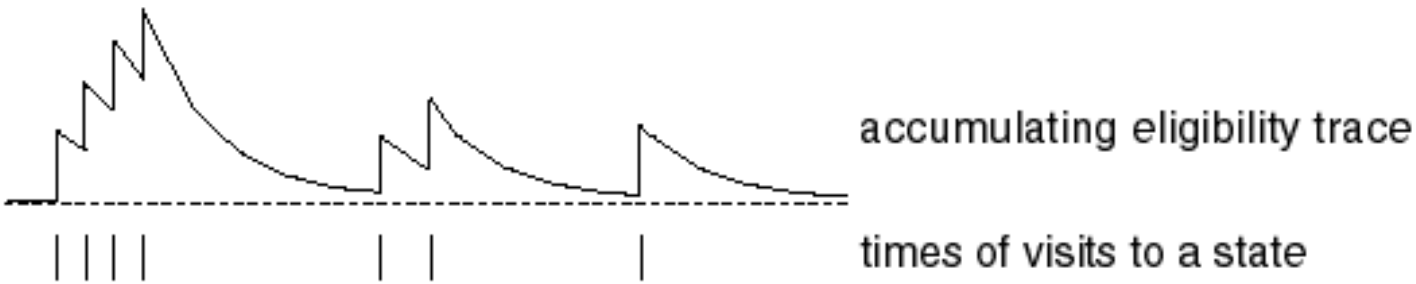

- Eligibility traces *keep a simple record of which states have recently been visited,* where "recently" is defined in terms of $\gamma\lambda$.

Backward - view TD(λ) : mechanism

- **TD error** : $\delta_t \; = \; R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

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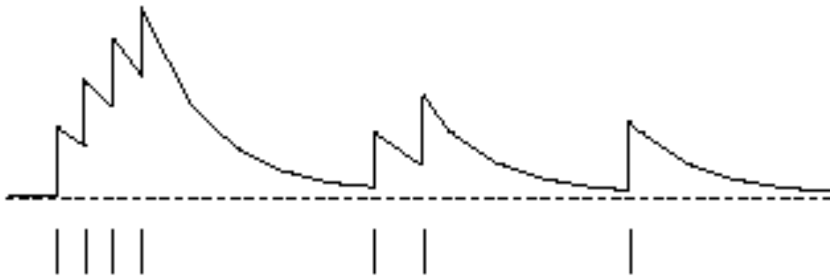
- **Eligibility traces update** : $E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$ **for all s**



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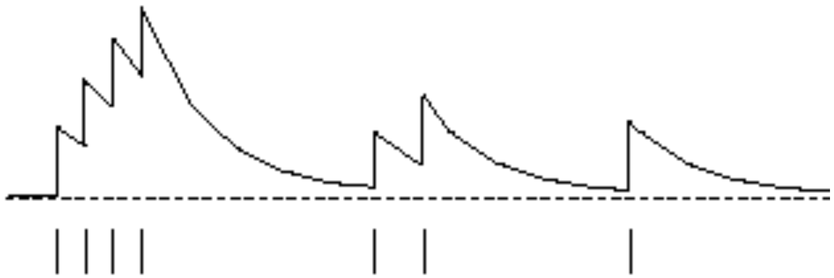
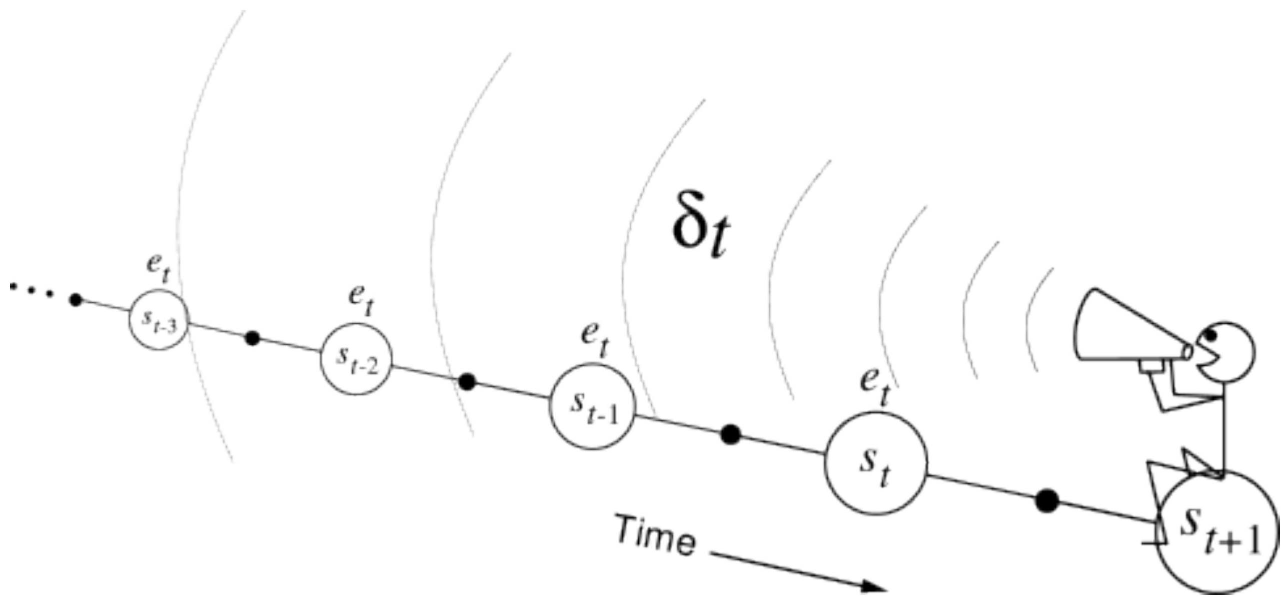


- **Value function update :** $V(s) \leftarrow V(s) + \alpha\delta_t E_t(s)$ **for all s**

Backward - view TD(λ) : mechanism

- **TD error** : $\delta_t \;=\; R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

- TD error is propagated backwards but in a decaying manner - *similar to the voice that fades away with the distance (the strength of the voice decreases with the temporal distance by $\gamma\lambda$)*

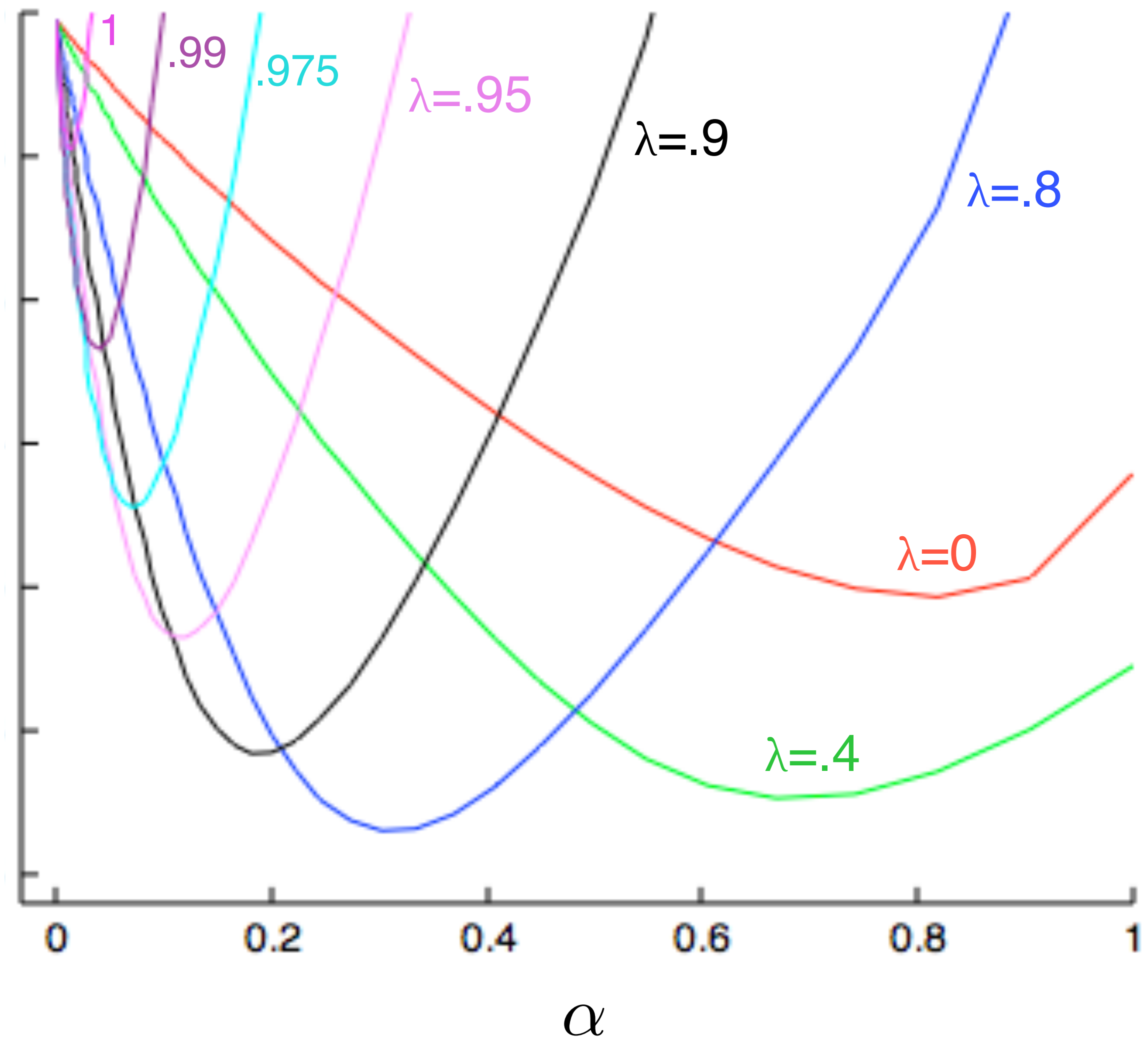


Backward - view TD(λ) : algorithm

Initialize $V(s)$ arbitrarily (but set to 0 if s is terminal)
Repeat (for each episode):
  Initialize $E(s) = 0$, for all $s \in \mathcal{S}$
  Initialize S
  Repeat (for each step of episode):
    $A \leftarrow$ action given by π for S
    Take action A, observe reward, R, and next state, S'
    $\delta \leftarrow R + \gamma V(S') - V(S)$
    $E(S) \leftarrow E(S) + 1$ (accumulating traces)
    For all $s \in \mathcal{S}$:
      $V(s) \leftarrow V(s) + \alpha \delta E(s)$
      $E(s) \leftarrow \gamma \lambda E(s)$
    $S \leftarrow S'$
  until S is terminal

Backward - view TD(λ) : algorithm

On-line TD(λ), accumulating traces



Backward – view TD(λ) and TD(0)

- When $\lambda = 0$, only current state is updated

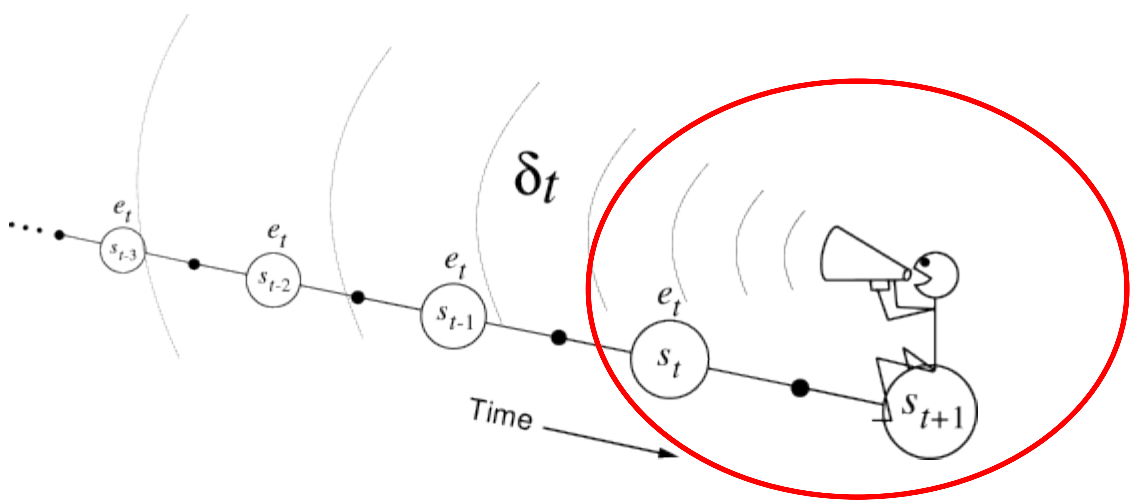
$$E_t(s) = \mathbf{1}(S_t = s)$$

$$V(s) \leftarrow V(s) + \alpha \delta_t E_t(s)$$

- This is exactly equivalent to TD(0) update

$$V(S_t) \leftarrow V(S_t) + \alpha \delta_t$$

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$



Equivalence of Forward and Backward views ?

Online updates

- TD(λ) updates are applied online at each step within episode

$$V(S) \leftarrow V(S) + \Delta V_t(S) \qquad \Delta V_t(S) = \alpha \delta_t E_t(s)$$

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Offline updates

- Updates are accumulated within episode
- but applied in batch at the end of episode

$$V(S) \leftarrow V(S) + \sum_{t=1}^{T} \Delta V_t(S)$$

Offline Equivalence of Forward and Backward views

Theorem

The sum of offline updates is identical for forward-view and and backward-view $TD(\lambda)$

$$\sum_{t=1}^{T} \alpha \delta_t E_t(s) = \sum_{t=1}^{T} \alpha \left(G_t^{\lambda} - V(S_t)\right) \mathbf{1}(S_t = s)$$

Offline Equivalence of TD(1) and MC

- Consider an episode where s is visited once at time-step k

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- Consider an episode where s is visited once at time-step k
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$$
\begin{aligned}
E_t(s) &= \gamma E_{t-1}(s) + \mathbf{1}(S_t = s) \\
&= \begin{cases} 0 & \text{if } \; t < k \\ \gamma^{t-k} & \text{if } \; t \geq k \end{cases}
\end{aligned}
$$

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- TD(1) updates accumulate error *online*

$$\sum_{t=1}^{T-1} \alpha \delta_t E_t(s) \longrightarrow V(s) \leftarrow V(s) + \sum_{t=1}^{T-1} \alpha \delta_t E_t(s)$$

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?

$$= \alpha \left(G_k - V(S_k) \right)$$

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$$

- By the end of the episode it accumulates total error

$$
\delta_k + \gamma \delta_{k+1} + \gamma^2 \delta_{k+2} + \ldots + \gamma^{T-1-k} \delta_{T-1}
$$

Telescoping in TD(1)

$$
\begin{aligned}
\delta_k + \gamma\delta_{k+1} + \gamma^2\delta_{k+2} + \ldots + \gamma^{T-1-k}\delta_{T-1} \\
&= R_{k+1} + \gamma V(S_{k+1}) - V(S_k) \\
&+ \gamma R_{k+2} + \gamma^2 V(S_{k+2}) - \gamma V(S_{k+1}) \\
&+ \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3}) - \gamma^2 V(S_{k+2}) \\
&\vdots \\
&+ \gamma^{T-1-k} R_T + \gamma^{T-k} V(S_T) - \gamma^{T-1-k} V(S_{T-1})
\end{aligned}
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Telescoping in TD(1)

$$\delta_k + \gamma\delta_{k+1} + \gamma^2\delta_{k+2} + \ldots + \gamma^{T-1-k}\delta_{T-1}$$

$$= R_{k+1} + \gamma \cancel{V(S_{k+1})} - V(S_k)$$

$$+ \gamma R_{k+2} + \gamma^2 V(S_{k+2}) - \gamma \cancel{V(S_{k+1})}$$

$$+ \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3}) - \gamma^2 V(S_{k+2})$$

$$\vdots$$

$$+ \gamma^{T-1-k} R_T + \gamma^{T-k} V(S_T) - \gamma^{T-1-k} V(S_{T-1})$$

Telescoping in TD(1)

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&+ \gamma^{T-1-k} R_T + \cancel{\gamma^{T-k} V(S_T)} - \cancel{\gamma^{T-1-k} V(S_{T-1})}
\end{aligned}$$

$$= R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \ldots + \gamma^{T-1-k} R_T - V(S_k)$$

$$= (G_k - V(S_k))$$

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- Backward TD(λ) updates accumulate error *online*

$$
\sum_{t=1}^{T} \alpha\delta_t E_t(s) = \alpha \sum_{t=k}^{T} (\gamma\lambda)^{t-k} \delta_t
$$

Forwards and Backwards TD(λ)

- Consider an episode where s is visited once at time-step k
- TD(λ) eligibility trace discounts time since visit,

$$E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$$
$$= \begin{cases} 0 & \text{if} \quad t < k \\ (\gamma\lambda)^{t-k} & \text{if} \quad t \geq k \end{cases}$$

- Backward TD(λ) updates accumulate error *online*

$$\sum_{t=1}^{T} \alpha\delta_t E_t(s) = \alpha \sum_{t=k}^{T} (\gamma\lambda)^{t-k} \delta_t \stackrel{?}{=} \alpha \left(G_k^\lambda - V(S_k)\right)$$

Telescoping in TD(λ)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\quad\vdots
\end{aligned}
$$

Telescoping in TD(λ)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+ (1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+ (1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\vdots
\end{aligned}
$$

$$
\begin{aligned}
&= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \lambda R_{k+1} - \lambda\gamma V(S_{k+1}) \\
&\quad + \lambda R_{k+1} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \lambda^2 R_{k+1} - \lambda^2\gamma R_{k+2} - \lambda^2\gamma^2 V(S_{k+2}) \\
&\quad + \lambda^2 R_{k+1} + \lambda^2\gamma R_{k+2} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots \\
&\quad \vdots
\end{aligned}
$$

Telescoping in TD(λ)

$$G_k^\lambda - V(S_k) = -V(S_k) + (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right)$$

$$+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right)$$

$$+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right)$$

$$\vdots$$

$$= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \lambda \cancel{R_{k+1}} - \lambda\gamma V(S_{k+1})$$

$$+\lambda \cancel{R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \lambda^2 R_{k+1} - \lambda^2\gamma R_{k+2} - \lambda^2\gamma^2 V(S_{k+2})$$

$$+\lambda^2 R_{k+1} + \lambda^2\gamma R_{k+2} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots$$

$$\vdots$$

Telescoping in TD(1)

$$G_k^\lambda - V(S_k) = -V(S_k) + (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right)$$

$$+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right)$$

$$+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right)$$

$$\vdots$$

$$= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \cancel{\lambda R_{k+1}} - \lambda\gamma V(S_{k+1})$$

$$+\cancel{\lambda R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \cancel{\lambda^2 R_{k+1}} - \lambda^2\gamma R_{k+2} - \lambda^2\gamma^2 V(S_{k+2})$$

$$+\cancel{\lambda^2 R_{k+1}} + \lambda^2\gamma R_{k+2} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots$$

$$\vdots$$

Telescoping in TD(1)

$$G_k^\lambda - V(S_k) = -V(S_k) + (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right)$$

$$+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right)$$

$$+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right)$$

$$\vdots$$

$$= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \lambda \cancel{R_{k+1}} - \lambda\gamma V(S_{k+1})$$

$$+\lambda \cancel{R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \lambda^2 \cancel{R_{k+1}} - \lambda^2 \cancel{\gamma R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2})$$

$$+\lambda^2 \cancel{R_{k+1}} + \lambda^2 \cancel{\gamma R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots$$

$$\vdots$$

Telescoping in TD(1)

$$G_k^\lambda - V(S_k) = -V(S_k) + (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right)$$

$$+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right)$$

$$+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right)$$

$$\vdots$$

$$= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \cancel{\lambda R_{k+1}} - \lambda\gamma V(S_{k+1})$$

$$+\cancel{\lambda R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \cancel{\lambda^2 R_{k+1}} - \cancel{\lambda^2\gamma R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2})$$

$$+\cancel{\lambda^2 R_{k+1}} + \cancel{\lambda^2\gamma R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots$$

$$\vdots$$

$$= R_{k+1} + \gamma V(S_{k+1}) - V(S_k) + \ldots$$

Telescoping in TD(1)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+(1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+(1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\vdots
\end{aligned}
$$

$$
\begin{aligned}
&= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \lambda \cancel{R_{k+1}} - \lambda\gamma V(S_{k+1}) \\
&\quad +\lambda \cancel{R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \lambda^2 \cancel{R_{k+1}} - \lambda^2\gamma \cancel{R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2}) \\
&\quad +\lambda^2 \cancel{R_{k+1}} + \lambda^2\gamma \cancel{R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots \\
&\qquad \vdots \\
&= R_{k+1} + \gamma V(S_{k+1}) - V(S_k) + \lambda\gamma\left(R_{k+2} + \gamma V(S_{k+2}) - V(S_{k+1})\right) + \ldots
\end{aligned}
$$

Telescoping in TD(1)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+ (1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+ (1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\vdots
\end{aligned}
$$

$$
\begin{aligned}
&= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \cancel{\lambda R_{k+1}} - \lambda\gamma V(S_{k+1}) \\
&\quad + \cancel{\lambda R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \cancel{\lambda^2 R_{k+1}} - \cancel{\lambda^2\gamma R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2}) \\
&\quad + \cancel{\lambda^2 R_{k+1}} + \cancel{\lambda^2\gamma R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots \\
&\qquad \vdots
\end{aligned}
$$

$$
= \underbrace{R_{k+1} + \gamma V(S_{k+1}) - V(S_k)}_{\delta_k} + \lambda\gamma\,(\underbrace{R_{k+2} + \gamma V(S_{k+2}) - V(S_{k+1})}_{\delta_{k+1}}) + \ldots
$$

Telescoping in TD(1)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+ (1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+ (1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\vdots
\end{aligned}
$$

$$
\begin{aligned}
= &-V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \cancel{\lambda R_{k+1}} - \lambda\gamma V(S_{k+1}) \\
&+ \cancel{\lambda R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \cancel{\lambda^2 R_{k+1}} - \cancel{\lambda^2\gamma R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2}) \\
&+ \cancel{\lambda^2 R_{k+1}} + \cancel{\lambda^2\gamma R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots \\
&\vdots
\end{aligned}
$$

$$
= \delta_k + \lambda\gamma\delta_{k+1} + \lambda^2\gamma^2\delta_{k+2} + \ldots
$$

Telescoping in TD(1)

$$
\begin{aligned}
G_k^\lambda - V(S_k) = -V(S_k) &+ (1-\lambda)\left(R_{k+1} + \gamma V(S_{k+1})\right) \\
&+ (1-\lambda)\lambda\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 V(S_{k+2})\right) \\
&+ (1-\lambda)\lambda^2\left(R_{k+1} + \gamma R_{k+2} + \gamma^2 R_{k+3} + \gamma^3 V(S_{k+3})\right) \\
&\quad \vdots
\end{aligned}
$$

$$
\begin{aligned}
&= -V(S_k) + R_{k+1} + \gamma V(S_{k+1}) - \cancel{\lambda R_{k+1}} - \lambda\gamma V(S_{k+1}) \\
&\quad + \cancel{\lambda R_{k+1}} + \lambda\gamma R_{k+2} + \lambda\gamma^2 V(S_{k+2}) - \cancel{\lambda^2 R_{k+1}} - \cancel{\lambda^2\gamma R_{k+2}} - \lambda^2\gamma^2 V(S_{k+2}) \\
&\quad + \cancel{\lambda^2 R_{k+1}} + \cancel{\lambda^2\gamma R_{k+2}} + \lambda^2\gamma^2 R_{k+3} + \lambda^2\gamma^3 V(S_{k+3}) + \ldots \\
&\quad \vdots
\end{aligned}
$$

$$
= \delta_k + \lambda\gamma\delta_{k+1} + \lambda^2\gamma^2\delta_{k+2} + \ldots \qquad = \sum_{t=k}^{T} (\gamma\lambda)^{t-k}\,\delta_t
$$

Forwards and Backwards TD(λ)

- Consider an episode where s is visited once at time-step k
- TD(λ) eligibility trace discounts time since visit,

$$E_t(s) = \gamma \lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$$
$$= \begin{cases} 0 & \text{if } \; t < k \\ (\gamma\lambda)^{t-k} & \text{if } \; t \geq k \end{cases}$$

- Backward TD(λ) updates accumulate error *online*

$$\sum_{t=1}^{T} \alpha \delta_t E_t(s) = \alpha \sum_{t=k}^{T} (\gamma\lambda)^{t-k} \delta_t = \alpha \left(G_k^\lambda - V(S_k) \right)$$

- By the end of the episode it accumulates total error for λ- return
- For multiple visits to s, $E_t(s)$ accumulates many errors

Online Equivalence of Forward and Backward views

- Forward and backward-view TD(λ) are slightly different

  the smaller α the closer the algorithms

Online Equivalence of Forward and Backward views

- Forward and backward-view TD(λ) are slightly different

  the smaller α the closer the algorithms

- NEW: Exact online TD(λ) achieves perfect equivalence
- By using a slightly different form of eligibility trace
- Sutton and von Seijen, ICML 2014

Summary of Forward and Backward TD(λ)

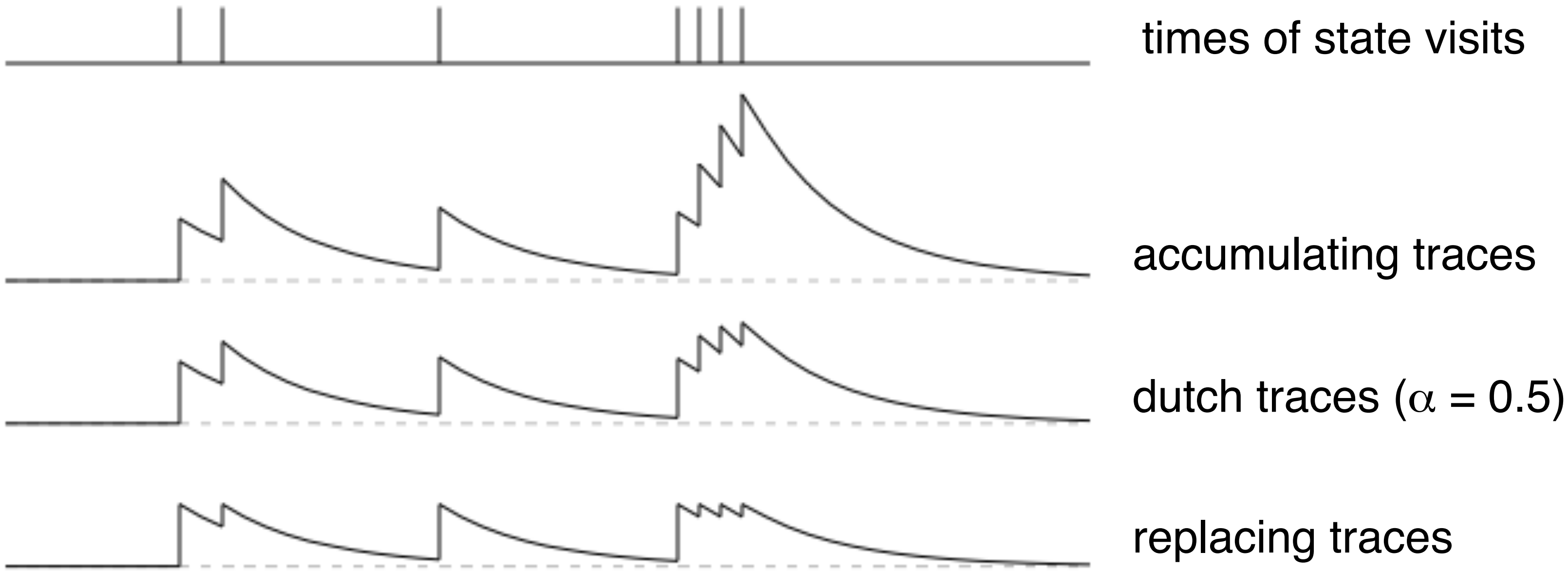
Offline updates	$\lambda = 0$	$\lambda \in (0, 1)$	$\lambda = 1$
Backward view Forward view	TD(0) $\parallel$ TD(0)	TD(λ) $\parallel$ Forward TD(λ)	TD(1) $\parallel$ MC
Online updates	$\lambda = 0$	$\lambda \in (0, 1)$	$\lambda = 1$
Backward view Forward view Exact Online	TD(0) $\parallel$ TD(0) $\parallel$ TD(0)	TD(λ) $\nparallel$ Forward TD(λ) $\parallel$ Exact Online TD(λ)	TD(1) $\nparallel$ MC $\parallel$ Exact Online TD(1)

$=$ here indicates equivalence in total update at end of episode.

Appendix : other definitions of eligibility traces

Initialize $V(s)$ arbitrarily (but set to 0 if s is terminal)
Repeat (for each episode):
    Initialize $E(s) = 0$, for all $s \in \mathcal{S}$
    Initialize S
    Repeat (for each step of episode):
        $A \leftarrow$ action given by π for S
        Take action A, observe reward, R, and next state, S'
        $\delta \leftarrow R + \gamma V(S') - V(S)$
        $E(S) \leftarrow E(S) + 1$ (accumulating traces)
        or $E(S) \leftarrow (1 - \alpha)E(S) + 1$ (dutch traces)
        or $E(S) \leftarrow 1$ (replacing traces)
        For all $s \in \mathcal{S}$:
            $V(s) \leftarrow V(s) + \alpha\delta E(s)$
            $E(s) \leftarrow \gamma\lambda E(s)$
        $S \leftarrow S'$
    until S is terminal

Appendix : other definitions of eligibility traces



- Accumulating traces add up each time a state is visited
- Replacing traces are reset to one
- Dutch traces do something inbetween, depending on α

In all cases the traces decay at a rate of $\gamma\lambda$ per step

Today's lecture…

- TD(λ) algorithm for model free prediction (as generalization of n-step return)
  - *Forward view*
  - *Backward view (Eligibility traces)*
  - *Equivalence between Forward and Backward views*

- Model free control : Sarsa (λ)

Model-free Control : n-Step Sarsa

- Consider the following n-step returns for $n = 1, 2, \infty$:

$$
\begin{array}{lll}
n = 1 & (\mathit{Sarsa}) & G_{t:t+1} = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) \\
n = 2 & & G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2}, A_{t+2}) \\
\vdots & & \\
n = \infty & (\mathit{MC}) & G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-1} R_{t+T}
\end{array}
$$

Model-free Control : n-Step Sarsa

- Consider the following *n*-step returns for $n = 1, 2, \infty$:

$$
\begin{aligned}
&n = 1 \quad (Sarsa) && G_{t:t+1} = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) \\
&n = 2 && G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2}, A_{t+2}) \\
&\quad\vdots && \\
&n = \infty \quad (MC) && G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-1} R_{t+T}
\end{aligned}
$$

- Define the *n*-step Q-return

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{n-1} R_{t+n} + \gamma^n Q(S_{t+n}, A_{t+n})$$

Model-free Control : n-Step Sarsa

- Consider the following *n*-step returns for $n = 1, 2, \infty$:

$$
\begin{array}{lll}
n = 1 & (Sarsa) & G_{t:t+1} = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) \\
n = 2 & & G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2}, A_{t+2}) \\
\vdots & & \\
n = \infty & (MC) & G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-1} R_{t+T}
\end{array}
$$

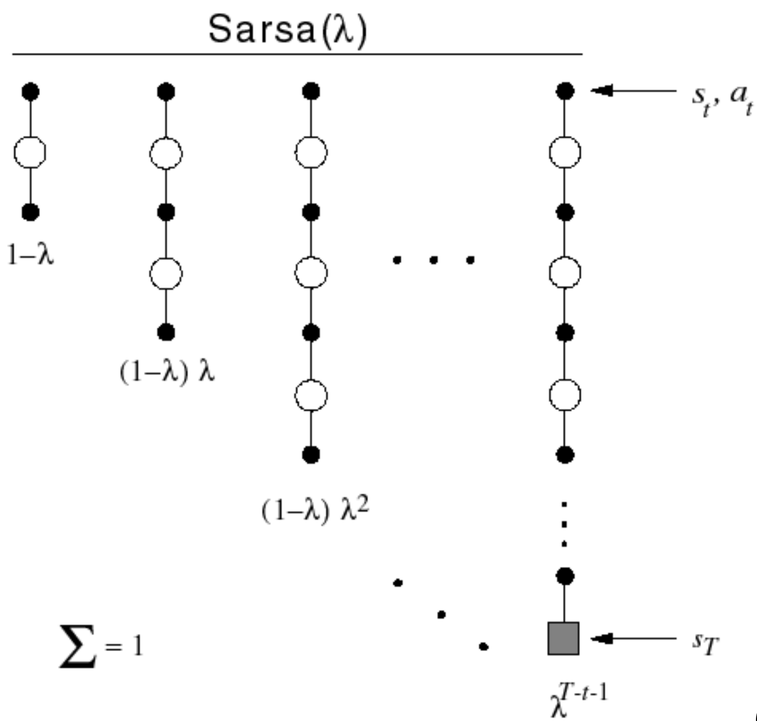
- Define the *n*-step Q-return

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{n-1} R_{t+n} + \gamma^n Q(S_{t+n}, A_{t+n})$$

- *n*-step Sarsa updates $Q(s, a)$ towards the *n*-step Q-return

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(G_{t:t+n} - Q(S_t, A_t))$$

Forward view Sarsa(λ)



- The q^λ *return* combines all n-step Q-returns $G_{t:t+n}$
- Using weight $(1-\lambda)\lambda^{n-1}$

$$q_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1}\, G_{t:t+n}$$

- Forward-view Sarsa(λ)

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left(q_t^\lambda - Q(S_t, A_t)\right)$$

Backward view Sarsa(λ)

- Just like TD(λ), we use eligibility traces in an online algorithm
- But Sarsa(λ) has one eligibility trace for each state-action pair

$$E_0(s, a) = 0$$
$$E_t(s, a) = \gamma \lambda E_{t-1}(s, a) + \mathbf{1}(S_t = s, A_t = a)$$

Backward view Sarsa(λ)

- Just like TD(λ), we use eligibility traces in an online algorithm
- But Sarsa(λ) has one eligibility trace for each state-action pair

$$E_0(s, a) = 0$$
$$E_t(s, a) = \gamma\lambda E_{t-1}(s, a) + \mathbf{1}(S_t = s, A_t = a)$$

- $Q(s, a)$ is updated for every state s and action a

Backward view Sarsa(λ)

- Just like TD(λ), we use eligibility traces in an online algorithm
- But Sarsa(λ) has one eligibility trace for each state-action pair

$$
\begin{aligned}
E_0(s, a) &= 0 \\
E_t(s, a) &= \gamma \lambda E_{t-1}(s, a) + \mathbf{1}(S_t = s, A_t = a)
\end{aligned}
$$

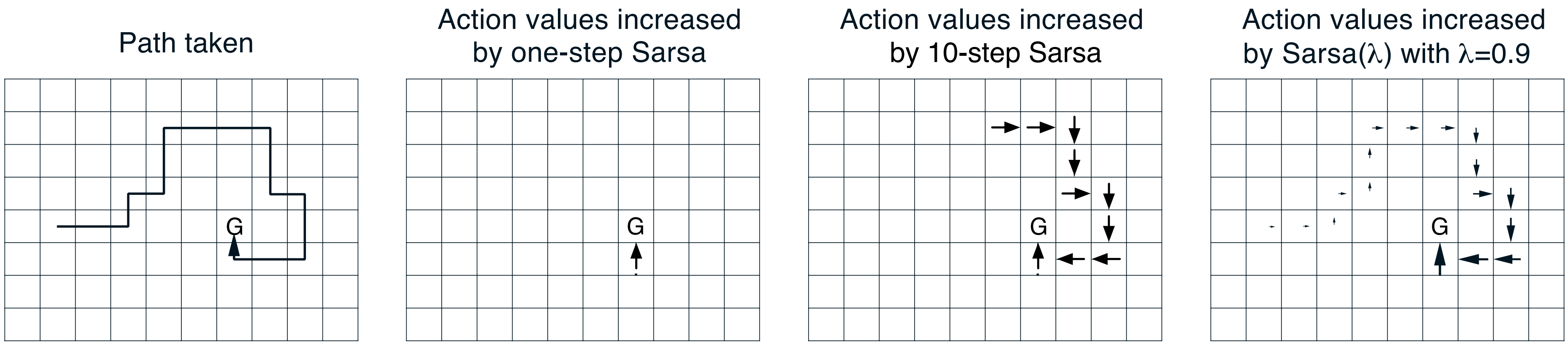
- $Q(s, a)$ is updated for every state s and action a
- In proportion to TD-error δ_t and eligibility trace $E_t(s, a)$

$$
\begin{aligned}
\delta_t &= R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t) \\
Q(s, a) &\leftarrow Q(s, a) + \alpha \delta_t E_t(s, a)
\end{aligned}
$$

Sarsa (λ) Algorithm

Initialize $Q(s,a)$ arbitrarily, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
Repeat (for each episode):
    $E(s,a) = 0$, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
    Initialize S, A
    Repeat (for each step of episode):
        Take action A, observe R, S'
        Choose A' from S' using policy derived from Q (e.g., ε-greedy)
        $\delta \leftarrow R + \gamma Q(S',A') - Q(S,A)$
        $E(S,A) \leftarrow E(S,A) + 1$
        For all $s \in \mathcal{S}, a \in \mathcal{A}(s)$:
            $Q(s,a) \leftarrow Q(s,a) + \alpha \delta E(s,a)$
            $E(s,a) \leftarrow \gamma \lambda E(s,a)$
        $S \leftarrow S'$; $A \leftarrow A'$
    until S is terminal

Sarsa (λ) Algorithm



The one-step method strengthens only the last action of the sequence

The *n*-step method strengthens the last n actions of the sequence

The trace method strengthens many actions of the sequence - *the degree of strengthening (indicated by the size of the arrows) falls off, according to $\gamma\lambda$ with steps from the reward.*

Material…

- The last version of the book described *eligibility traces* in Chapter 12…
- …but it is mixed with value function approximation algorithms (in my opinion it is a bit confusing)
- A previous *in progress* version of the book (2014/1015) describes in Chapter 7 (*more and less*) the material I have presented. I will upload this *in progress* version of he book.
- *More and less* means that sometimes the notation is slightly different and that I have presented just a part of Chapter 7 of the *in progress* version 2014/2015; in particular I just mention Dutch traces and I did not discuss Off-policy importance sampling algorithms
- The material required for the final exam is the one I have presented today in the slides (please read Chapter 7 of the *in progress* version 2014/2015 as reference for intuition and deep explanations)